# Utku Kalkanli

AI Engineer

Phone: +49 172 7948165 | Email: kalkanliutku@gmail.com | Munich, Germany

linkedin.com/in/utku-kalkanli | github.com/utkukalkanli | utkukalkanli.github.io

**Available immediately · Full-time**

AI engineer with three years at Linde building LLM, RAG, GraphRAG, and agent systems and running ML in production on Azure.

## Experience

**Linde GmbH, Digitalization** Oct 2023 – Sep 2026
*Machine Learning Engineer* Munich, Germany

- Built the document ingestion and retrieval pipeline of Linde's internal document-processing agent, used by ∼50,000 employees (chat with uploaded documents, Excel and SharePoint content; report generation): Azure OpenAI, LangChain, semantic chunking, Qdrant hybrid search and reranking; evaluated with RAGAS (faithfulness, context precision/recall, response relevancy, factual correctness).
- Developed a Neo4j GraphRAG workflow that brings connected enterprise knowledge into retrieval and answer generation.
- Maintained and optimized a production Azure ML pipeline that forecasts CPAP therapy adherence for ∼1M patients daily. Cut runtime from 37 to 8 hours (78%) and added data-quality and drift monitoring.
- Delivered an RF-DETR small-object detector for satellite imagery (mAP@50 0.66 → 0.83) with an Airflow pipeline for training, TensorRT export, and Triton evaluation; deployed for the sales team.
- Contributed to a rotor-orbit fault classifier deployed at a Linde air separation plant, where it predicted a real equipment failure. Named inventor on the invention; co-author of an internal predictive-maintenance whitepaper.

**Fraunhofer IKS** Oct 2025 – Oct 2026
*Research Assistant, Machine Learning (Master's Thesis)* Munich, Germany

- Building an autonomous LangGraph agent with Qwen tool calling that orchestrates multi-stage PyTorch experiments, domain adaptation, and evaluation, reducing manual experiment setup.
- Thesis: CAD-referenced anomaly detection with CLIP, DINOv2, VGGT, and diffusion-generated training data; AUROC 0.73 on real photographs without real training images.

**Novit.AI** Mar 2022 – Jun 2022
*Machine Learning Engineer, Internship* Ankara, Türkiye

- Built and deployed a containerized edge AI system (FastAPI, Docker, XGBoost, GitHub Actions, NGINX) with real-time inference and automated rollout to 50+ NVIDIA Jetson devices.

**Open Business Software Solutions; Cybersoft** 2020 – 2021
*Software Engineer, Internships* Türkiye

- Built microservices and web apps with Python, Django, PostgreSQL, Elasticsearch, React and TypeScript on GCP.

## Education

**Technical University of Munich (TUM)** Apr 2023 – Oct 2026
*M.Sc. in Informatics* Munich, Germany
Focus: machine learning, databases, high-performance computing. Thesis: *Reference-Guided Multi-view Anomaly Detection for Industrial Objects* (Prof. Stephan Günnemann, at Fraunhofer IKS).

**Bilkent University** Sep 2017 – Jun 2022
*B.Sc. in Computer Science* Ankara, Türkiye

## Projects

**Radar-Based Ship Identification** (TUM industry project with Covadonga Maritime, Nov 2024 – Jun 2025)
Built an automated labeling pipeline matching radar sightings to AIS ship tracks (240 s temporal clustering; one year, 935k sightings, 388 ships), radar-signal features (peak detection, FFT, autocorrelation), and an XGBoost classifier: 91% accuracy on a class-balanced test set. Delivered to Covadonga.

## Skills

**LLM and Agents:** LLMs, Azure OpenAI, LangChain, LangGraph, LlamaIndex, Pydantic, MCP, Tool Calling, Prompt Engineering, LLM Evaluation (RAGAS), Qwen, Hugging Face
**RAG:** RAG, GraphRAG, Semantic Chunking, Hybrid Retrieval, Vector Search (Qdrant), Reranking, Neo4j
**Cloud and MLOps:** Microsoft Azure, Azure Machine Learning (Azure ML), Azure DevOps, GCP, MLOps, Apache Airflow, MLflow, Docker, CI/CD, GitLab, Ray, Slurm
**APIs and Serving:** Python, TypeScript, SQL, FastAPI, REST APIs, Microservices, PyTorch, scikit-learn, ONNX, TensorRT, NVIDIA Triton, AWS (EC2, S3)
**Computer Vision:** YOLO, RF-DETR, CLIP, DINOv2, VGGT, Anomaly Detection, 3D Reconstruction, Pose Estimation
**AI-Assisted Development:** Claude Code, Codex
**Languages:** Turkish (native), English (C1, fluent), German (A2)